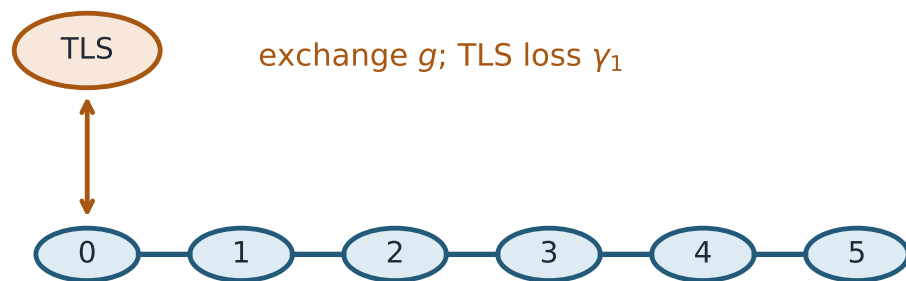


a Finite chain and local lossy probe



TLS contact x is movable

b Common physical preparation

$$\rho(0) = |\uparrow_z\rangle\langle\uparrow_z|^{\otimes N} \otimes |0_d\rangle\langle 0_d|$$

Same initial state for all principal comparisons

No Floquet-pair or topology-selected preparation

c Frequency-resolved TLS coherence

$$c_d(t) = [\langle X_d \rangle + i\langle Y_d \rangle]/2$$



$$A_{\text{TLS}}(r) \text{ and } W_r = \int dr A_{\text{TLS}}^2(r)$$

$r = \omega_d/(\Omega/2)$, with a declared detector grid

d Questions answered by the controls

OBC/PBC and contact position: boundary selectivity

Matched drives: limits of the ν_π label

Fixed eight-drive $N = 4 \rightarrow 6$: weight transfer